

2(a)	<i>any three from</i> 1 (diagram shows) hydrogen bond is a weak bond ; 2 each oxygen (atom) forms two hydrogen bonds / each hydrogen (atom) forms one hydrogen bond ; 3 (attraction) between oxygen (atom) of one water molecule and hydrogen (atom) of another (forms a hydrogen bond) ; R cohesion / adhesion – <i>if used for attraction</i> 4 water is <u>dipolar</u> ; 5 detail ; e.g. electrons not shared equally between oxygen and hydrogen A oxygen is more electronegative (than hydrogen) A <i>ref. to</i> (two) lone pair(s) (of electrons) on oxygen A uneven distribution of, electrons / charge e.g. oxygen has, <u>small</u> / <u>slight</u>, negative charge / δ^-, and, hydrogen has, <u>small</u> / <u>slight</u>, positive charge / δ^+ <i>only needs to state 'small' once</i>	3
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Question	Answer	Marks
2(b)(i)	<i>idea that</i> (H-bonds) maintain / AW, (shape / structure, form of) α-helices / β-pleated sheets ; A allows formation of, α-helices / β-pleated sheets R if bonds are between R groups	1
2(b)(ii)	<i>any two from</i> 1 <i>idea that</i> hydrogen bonds help to, stabilise / AW, further folding of, amylase / polypeptide / protein ; 2 between, R groups with amine and carboxyl groups ; A between R groups with -NH and, -CO / -OH 3 <i>idea that</i> may be between amino acids far apart in primary structure ; 4 <i>either</i> helps to maintain / form / AW, globular shape / 3D shape / structure (of amylase / polypeptide / protein) or maintains / forms / AW, (specific) shape / structure, of, active site / binding site ;	2
2(c)	<i>any three from</i> 1 dissolves / AW, ions / minerals / salts, <u>and</u> (named) polar molecules ; A 'assimilates' as polar I substances / nutrients 2 transports, solute(s) / named solutes / dissolved substance, in, xylem / phloem / xylem and phloem ; 3 storage of, solutes / named solutes, in vacuoles ; 4 metabolic / chemical / cellular, reactions occur in water ; 5 dissolves, carbon dioxide / oxygen, with ref to, respiration / photosynthesis;	3